Supplementary Figure 16	Comparisons of the tracking results of B-cell dynamics obtained by CELLECT and Imaris
Supplementary Figure 17	Extracted traces of highly migratory B cells
Supplementary Figure 18	Tracking performance of CELLECT on terabyte-level imaging datasets of B-cell dynamics
Supplementary Figure 19	Influence of the intensity threshold on the segmentation results of Imaris with comparisons to CELLECT
Supplementary Figure 20	Continuous tracking and segmentation of a single neutrophil chasing and engulfing bacteria

II. Supplementary Tables

Supplementary Table 1	Quantitative results on mskcc-confocal and nih-ls data
Supplementary Table 2	Quantitative results for each fold in the cross-validation experiments on mskcc-confocal and nih-ls data
Supplementary Table 3	Input and output tensor dimensions for each module
Supplementary Table 4	Quantitative ablation study of individual modules in the CELLECT framework

III. Supplementary Videos

Supplementary Video 1	Confidence map generated by CELLECT on a dataset of membrane-labeled cells
Supplementary Video 2	Efficient cell tracking of CELLECT on *C. elegans* dataset compared with Linajea
Supplementary Video 3	3D cell tracking of large-scale B cells during the formation of germinal center
Supplementary Video 4	Continuous tracking of neural activities at single-cell resolution in *Drosophila* brain during tissue deformation